The Frontend Journey: From HTML to Intelligent Interfaces

01

This Presentation isn’t going to be about definitions.

02

It isn’t gonna be about History of the programming languages and frameworks or thier evolution or anything theoretical.

03

It is going to be about demonstation and about how you can go about building entire frontend projects on your own (Not with AI or tools like wordpress)

How HTML works

- Not a programming language - A markup language
- Used only to define structure, not logic or perform calculations.
- Declarative – Doesn’t need details
- Browser Interprets HTML (HyperText Markup Language and Creates DOM

HTML Demo

HTML already provides interaction before we write a single line of JavaScript.

It can Navigate Links

It can Submit Forms

It can validate inputs

Form and Accessibilty Demo

What is HTTPBin

- It does not store your data permanently.

- It just “echoes” it back in the response.

- You can find same data in the browser devtool’s response window.

- It is a small web service designed for:

  - Testing HTTP requests
  - Debugging APIs
  - Learning how HTTP works
  - Demonstrating form submissions

Styling the Web: How CSS Transforms HTML

HTML builds the structure.
CSS tells the browser how to paint it.

- Browser parses CSS → Builds CSSOM
- DOM + CSSOM → Render Tree
- Render Tree → Layout → Paint → Pixels

CSS Demo

https://github.com/saurabveersingh/birthday-card

https://github.com/saurabveersingh/business-card

https://mvup.leanpitch.com/

From Static to Interactive: Enter JavaScript

Alternative: Typescript

01 The engine behind modern web applications

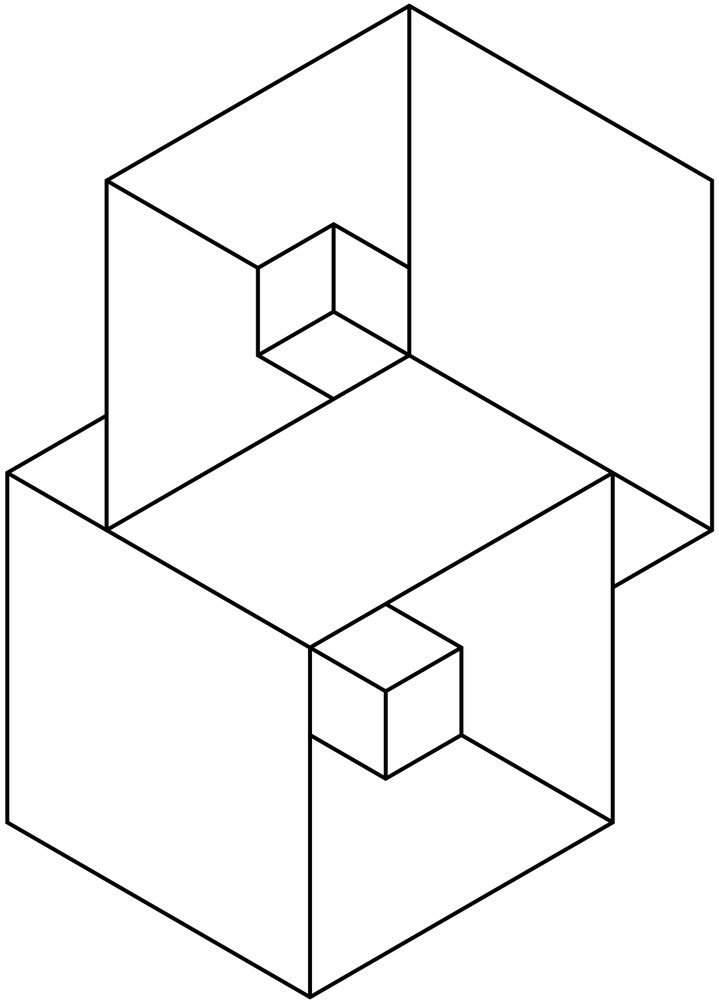
02 Responsible for logic and interactivity

03 Can Read and Modify the DOM

04 Listen to events

05 Fetch Data

06 Update UI dynamically



JavaScript Demo

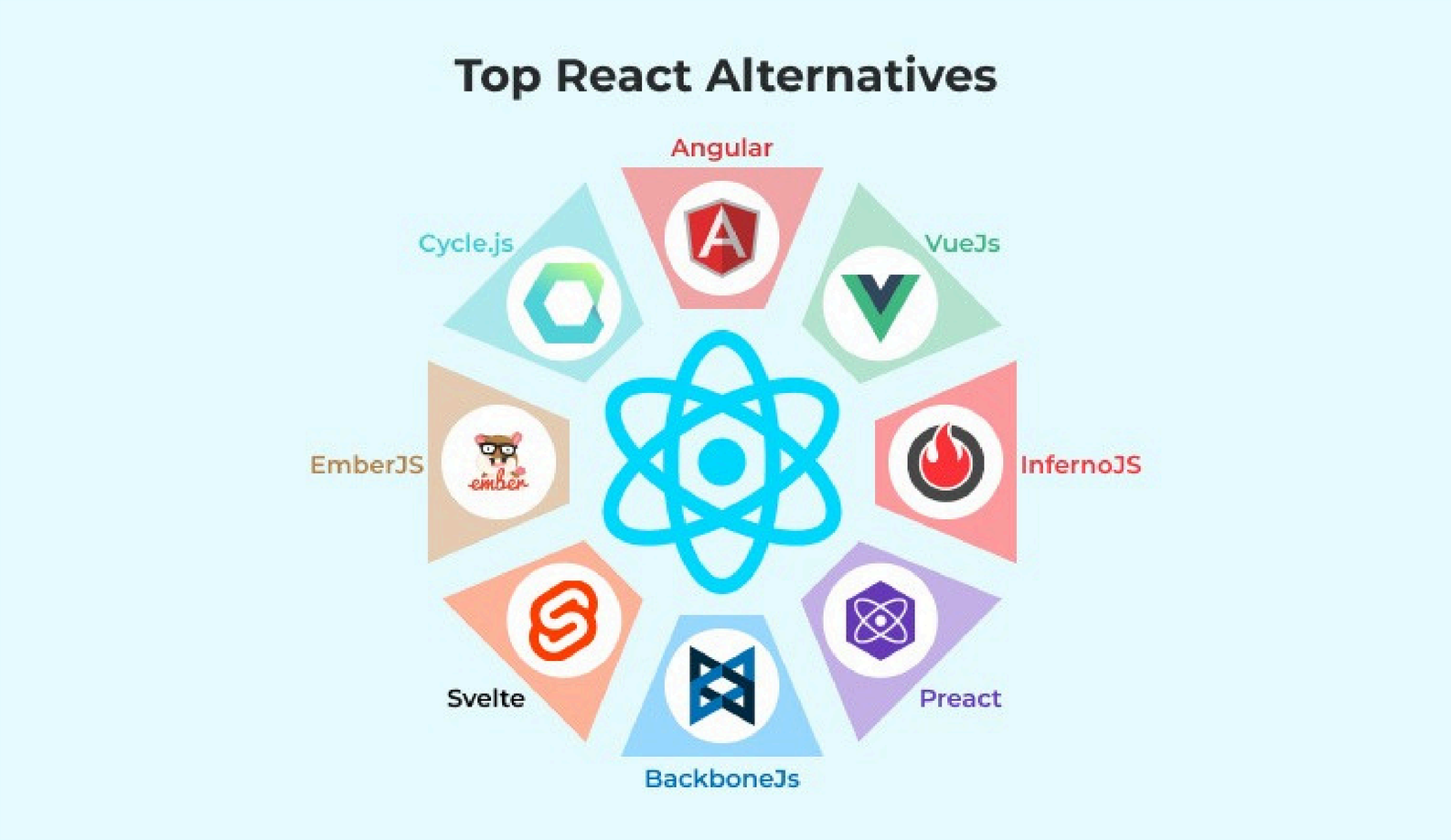
https://github.com/saurabveersingh/Cats_meme_picker

Scaling Interactivity with React

Declarative Programming

Uses JSX to describe Markup within JavaScript files

Component Based

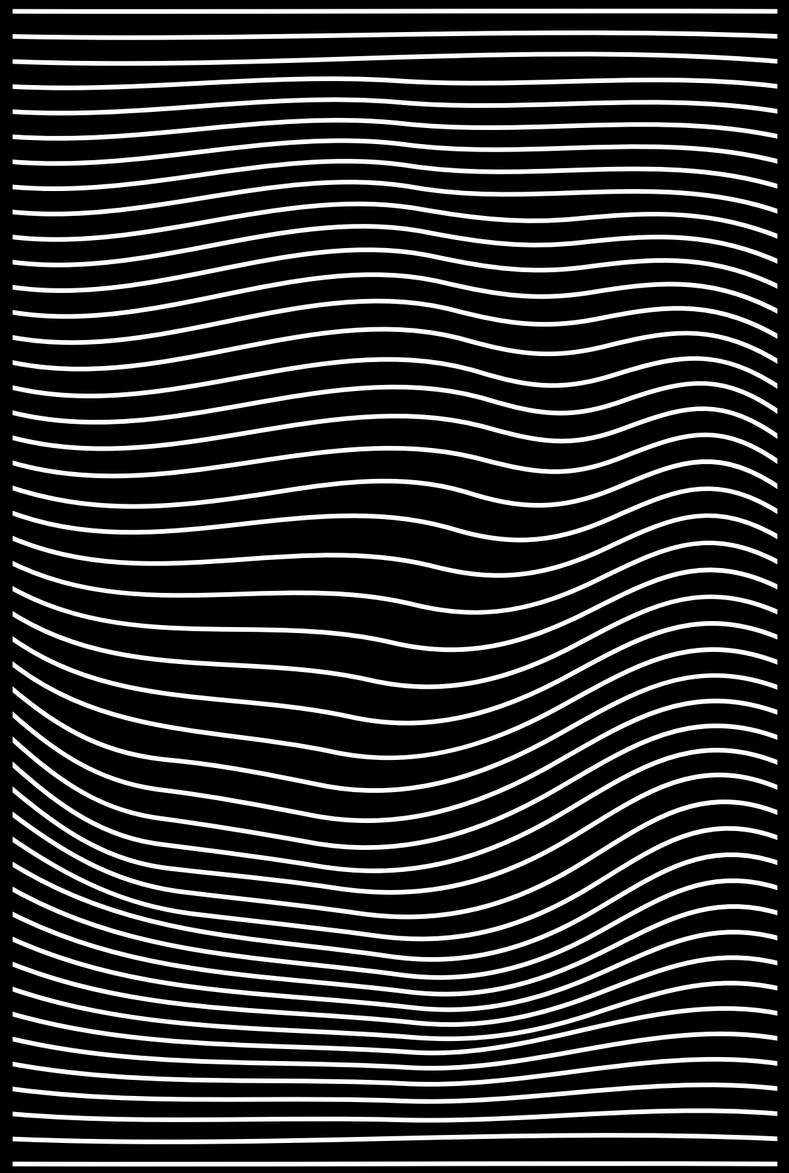

Top React Alternatives
Angular
Cycle.js
VueJs
EmberJS
InfernoJS
Svelte
Preact
BackboneJs

Basic Features in React for powerful Apps

- JSX – Conditional rendering, events
- State
- Effects
- Components
- Props
- Router

React Demo

https://github.com/saurabveersingh/Form-Response



Advanced Features in React

Custom Hooks

Context API to Share data across components without prop drilling

Lazy Loading

Advanced React Demo

https://retina-vault.netlify.app/

How does Next.js add Value

- Server side Rendering
- Automates lazy loading
- SEO optimization
- File-Based Routing - Zero Config
- built-in backend support

Questions?

Thank You